

Converters:

- ① A D/A converter has 6 bits and a reference voltage of 10V. Calculate the minimum value of R such that the maximum value of o/p current does not exceed 10mA. Find also the smallest quantized value of o/p current.

⇒ we have:

$$V_o = V_{ref} \left(-\frac{R_f}{R} \right) \left[1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} \right]$$

$$= V_{ref} \left(-\frac{R_f}{R} \right) \left[\frac{1 \cdot (1 - \frac{1}{2^n})}{1 - \frac{1}{2}} \right] \left[\because S_n = \frac{a(1-r^n)}{1-r} \text{ for } r < 1 \right]$$

$$= V_{ref} \left(-\frac{R_f}{R} \right) \left(\frac{2^n - 1}{2^n} \times 2 \right)$$

$$= V_{ref} \left(-\frac{R_f}{R} \right) \left[\frac{2^n - 1}{2^{n-1}} \right]$$

For max^m current, all the bits = '1', So the value of all switches = 1 & ~~then~~ $n=6$

$$\frac{V_o}{R_f} = \frac{V_{ref}}{R} \times \frac{2^6 - 1}{2^5}$$

$$\text{or, } I_{\max} = \frac{10}{R} \times \frac{63}{32}$$

$$\text{or, } R = \frac{10}{10 \times 10^{-3}} \times \frac{63}{32}$$

$$\therefore R = 2000 \Omega$$

Now, Smallest quantized current = Current with LSB

$$\therefore (I_o)_{\text{LSB}} = \frac{V_{ref}}{R} \times \frac{1}{2^{n-1}} = \frac{10}{2k} \times \frac{1}{32} = 156 \mu A$$

- ② Consider a 6-bit D/A Converter with a resistance of $320k\Omega$ in LSB position. The converter is designed with weighted resistive network. The reference voltage is $10V$. The o/p of the resistive n/w is connected to an op-Amp with a feedback resistance of $5k\Omega$. What is the o/p voltage for a binary i/p of 111010?

⇒ Here,

$$V_o = V_{ref} \left(\frac{R_f}{R} \right) \left[\frac{S_1}{2^1} + \frac{S_2}{2^2} + \frac{S_3}{2^3} + \frac{S_4}{2^4} + \frac{S_5}{2^5} + \frac{S_6}{2^6} \right]$$

$$\text{or } I_o = \frac{V_o}{R_f} = \frac{V_{ref}}{R} \left[\frac{S_1}{2} + \frac{S_2}{2^2} + \frac{S_3}{2^3} + \frac{S_4}{2^4} + \frac{S_5}{2^5} + \frac{S_6}{2^6} \right] \quad \text{--- (1)}$$

and resistance at LSB position $= 2^{n-1} R$

$$320k = 2^{6-1} R$$

$$\therefore R = 10k\Omega$$

Now, from eqⁿ (1),

$$I_o = \frac{10}{10 \times 1000} \left[1 + 0.5 + 0.25 + 0 + 0.0625 + 0 \right]$$

$$= 10^{-3} \times 1.8125$$

$$= 1.8125mA$$

$$\text{The o/p voltage} = I_o \times R_f$$

$$= 1.812m \times 5k$$

$$= 9.0625V$$

$$\therefore E_o = -9V \quad [\because \text{inverting confⁿ of op-Amp}]$$

- ③ Find the Successive approximation A/D output for a 4-bit converter to a 3.217V i/p if the reference voltage is 5V.
 $\Rightarrow$ The o/p of DAC is given by

$$V_o = V_{ref} \left[\frac{d_0 \times 2^0 + d_1 \times 2^1 + d_2 \times 2^2 + d_3 \times 2^3}{2^4} \right]$$

$$= \frac{V_{ref}}{16} [d_0 + 2d_1 + 4d_2 + 8d_3] \quad \text{--- (1)}$$

If digital o/p be $d_3 d_2 d_1 d_0$ then,

(i) Set $d_3 = 1$,

$$V_o = \frac{5}{16} [0 + 0 + 0 + 8] = 2.5V$$

Here $3.217 > 2.5 \therefore d_3 = 1$

(ii) Set $d_2 = 1$

$$V_o = \frac{5}{16} [0 + 0 + 4 + 8] = 3.75V$$

Here $3.217 < 3.75, \therefore d_2 = 0$

(iii) Set $d_1 = 1$

$$V_o = \frac{5}{16} (0 + 2 + 0 + 8) = 3.125V$$

Here, $3.217 > 3.125, \therefore d_1 = 1$

(iv) Set $d_0 = 1$

$$V_o = \frac{5}{16} (1 + 2 + 0 + 8) = 3.4375V$$

Here $3.217 < 3.4375, \therefore d_0 = 0$

Hence the digital o/p = $d_3 d_2 d_1 d_0 = 1010$